Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Graduate **Software Engineer** with 3 years of industry experience building and deploying **Python** services, **ReactJS** frontends and **data pipelines** in regulated financial services, and an **MSc in Data Science** from the University of Bristol (awaiting result). Hands-on **computer vision** project work: a multi-architecture image-captioning study (**CNN, ViT and ResNet** encoders) and a **CNN**-trained self-driving car in the Udacity simulator. Ran a self-hosted LLM on a **GPU** instance, works directly with stakeholders and acted as de facto team lead for up to 4 interns; twice recognised with Best Team Project. Keen to apply this foundation to real-time media and video systems and to build **C++** skills with mentoring.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Software Engineering & Systems

- Designed and built **production applications** end to end: **ReactJS** frontend served by **Nginx**, **Python FastAPI** microservices in **Docker** (uvicorn) on **AWS EC2** and a **PostgreSQL** (pgvector) database; initially the only engineer alongside my manager, later working with a frontend developer and leading up to 4 interns as de facto team lead.
- Automated delivery of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins and GitLab CI/CD** pipeline; secured the apps with **RBAC** (Python **LDAP** authentication), AWS Security Groups limited to company private IPs and an **Nginx** reverse proxy, one of the fixes that took them from repeatedly failing to passing the security team's **VAPT**.
- Built **RAG** and tool-calling applications with **LlamaIndex** and **Azure OpenAI GPT-3.5 & Llama-3**, with a separate **pgvector** database per department; ran **self-hosted Llama-3 on a GPU EC2 instance** for the largest department to control cost, with the ability to switch models.

- Worked directly with Client Service, Compliance and Investment stakeholders to turn requests into requirements and their feedback into releases; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

Data Processing & Analysis

- Built a **Python** pipeline, scheduled with EventBridge, Lambda and AWS SageMaker, that parses over **300 daily emails** from credit rating agencies into structured data and fuzzy-matches company names, **saving 60 analyst hours a month**; replaced AWS Textract with Azure OpenAI when a rating agency began sending emails as images.
- Designed a **Normalized SQL Schema** (PostgreSQL) and wrote **Python ingestion scripts** for the Morningstar API covering every fund in the Indian mutual fund market (2023–2025), integrated with legacy analyst data; developed **Optimized SQL** queries, Functions and Procedures.

- Analyzed customer data with **Python Pandas** and Market Basket Analysis (**Apriori Algorithm**), handing the insights to the marketing team; migrated analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**.

PROJECTS

- **University of Bristol MSc Applied Data Science Group Project: VisionCaptioner** (Team project, 5 authors. Computer vision study generating natural-language descriptions of synthetic images (tic-tac-toe boards, multi-digit numbers, coloured shapes, 10,000 images each): SimpleCNN, DeeperCNN, ResNet-18 and custom ViT encoders trained from scratch with symbolic, SBERT embedding-retrieval and LSTM sequence decoders (with and without Bahdanau attention), run over five seeds, scored with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca: CellLineSelector** (Team project, co-built. Data preprocessing pipeline joining multiple raw datasets into clean tables with integrity checks, scoring algorithm and RAG-based AI assistant for a tool, built as FastAPI microservices with a React frontend, that ranks cancer cell lines by target inclusion and exclusion genes using per-gene scores across six evidence layers; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **B.Tech Final Year Project:** Self-driving car trained using a Convolutional Neural Network (CNN) model in the Udacity self-driving car simulator.
- **Wingman:** AI dating-coach web app (ReactJS, Python FastAPI) with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos; LLM output treated as untrusted (validated with Pydantic, cached by input hash) and results streamed to the UI to work within rate limits. GitHub: https://github.com/Arishtanemi4/wingman.git

ACHIEVEMENTS

- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Computer Vision:** CNN, ViT and ResNet architectures; image captioning (VisionCaptioner); CNN-based self-driving car (Udacity simulator).
- **Tools:** Git, Docker, Jenkins, GitLab (CI/CD), Linux, AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Azure OpenAI, LlamaIndex, Nginx, LDAP, Power BI, Claude Code (AI-assisted development), VS Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite